

CSA 0921 - Programming in Java

CO2 - AT2 - ASSIGNMENT 1

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The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo and the text "SIMATS Online Programming Lab" and "JAVA PROGRAMMING". The user's name "Vidhushini T S" and registration number "192511061" are displayed. The assignment title "CO2-AT2-ASSIGNMENT 1" is shown. The question is "Q1 Multi-Specialty Hospital Patient Record Management" with 50 marks. The description states: "A multi-specialty hospital maintains patient records collected from emergency, outpatient, inpatient, laboratory, and pharmacy departments. Since different departments independently register patients, duplicate patient IDs, inconsistent patient names, repeated addresses, and multiple phone numbers often exist in the database. Hospital management wants a centralized application capable of cleaning patient records and organizing them efficiently for quick retrieval during emergencies. The software should utilize Java String Handling techniques and Collection Framework classes to maintain accurate records. Develop a Java application to perform the following operations: a) Read Patient ID, Name, Age, Gender, Department, Blood Group, Contact Number, and Diagnosis. b) Clean patient names and addresses using appropriate String methods. c) Store patient objects using an ArrayList. d) Remove duplicate Patient IDs using a HashSet. e) Maintain a HashMap that maps Patient IDs with diagnosis details. f) Display all patient records using Iterator. g) Implement the application using Generics. h) Produce department-wise patient reports and discuss the advantages of using different Collection Interfaces." The "Your Code" section contains the following Java code:

```
1 import java.util.*;
2 //Generic Patient Class(g) implement the application using Generics
3 class Patient<T>{
4     private T id;
5     private String name;
6     private int age;
7     private String gender;
8     private String department;
9     private String bloodGroup;
10    private String contactNumber;
11    private String diagnosis;
12    public Patient(T id, String name,int age, String gender,String dep
13    this.id=id;
14    // (b) Clean patient names using appropriate string methods(trim
15    this.name=cleanString(name);
16    this.age=age;
17    this.gender=gender.trim();
18    this.department=department.trim();
19    this.bloodGroup=bloodGroup.trim();
20    this.contactNumber=contactNumber.trim();
21    this.diagnosis=diagnosis.trim();
22 }
23 private String cleanString(String input){
24     if(input==null||input.trim().isEmpty()) return "";
25     String[] words=input.trim().replaceAll("\\s+"," ").split(" ");
26     StringBuilder sb=new StringBuilder();
27     for(String word:words){
28         if(word.length()>0){
29             sb.append(word.substring(0,1).toUpperCase());
30             sb.append(word.substring(1).toLowerCase());
31             sb.append(" ");
32         }
33     }
34     return sb.toString().trim();
35 }
36 public T getId(){return id;}
37 public String getName(){return name;}
```

The "Input" section contains the following text:

```
2
P101
John doe
34
Male
Cardiology
0+
9876543210
Hypertension
P101
John doe
34
Male
Cardiology
0+
9876543210
Hypertension
P102
Jane smith
29
Female
Neurology
A+
9123456789
Migraine
```

The screenshot shows the SIMATS Online Programming Lab interface with the completed code and output. The "Your Code" section contains the following Java code:

```
1 import java.util.*;
2 //Generic Patient Class(g) implement the application using Generics
3 class Patient<T>{
4     private T id;
5     private String name;
6     private int age;
7     private String gender;
8     private String department;
9     private String bloodGroup;
10    private String contactNumber;
11    private String diagnosis;
12    public Patient(T id, String name,int age, String gender,String dep
13    this.id=id;
14    // (b) Clean patient names using appropriate string methods(trim
15    this.name=cleanString(name);
16    this.age=age;
17    this.gender=gender.trim();
18    this.department=department.trim();
19    this.bloodGroup=bloodGroup.trim();
20    this.contactNumber=contactNumber.trim();
21    this.diagnosis=diagnosis.trim();
22 }
23 private String cleanString(String input){
24     if(input==null||input.trim().isEmpty()) return "";
25     String[] words=input.trim().replaceAll("\\s+"," ").split(" ");
26     StringBuilder sb=new StringBuilder();
27     for(String word:words){
28         if(word.length()>0){
29             sb.append(word.substring(0,1).toUpperCase());
30             sb.append(word.substring(1).toLowerCase());
31             sb.append(" ");
32         }
33     }
34     return sb.toString().trim();
35 }
36 public T getId(){return id;}
37 public String getName(){return name;}
```

The "Output" section contains the following text:

```
ALL UNIQUE PATIENT RECORDS
ID: P101|Name: John
Doe|Age:34|Gender:Male|Dept:Cardi
ology|Blood
Group:9876543210|ContactNo:9123456789
```

SIMATS Online Lab

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SIMATS Online Programming Lab

JAVA PROGRAMMING

Vidhushini T S 1923110619C

```
36 public T getId(){return id;}
37 public String getName(){return name;}
38 public int getAge(){return age;}
39 public String getGender(){return gender;}
40 public String getDepartment(){return department;}
41 public String getBloodGroup(){return bloodGroup;}
42 public String getContactNumber(){return contactNumber;}
43 public String getDiagnosis(){return diagnosis;}
44 @Override
45 public String toString(){
46     return "ID: "+id+"Name: "+name+"Age:"+age+"Gender:"+gender+"
47 }
48 @Override
49 public boolean equals(Object o){
50     if(this==o) return true;
51     if(o==null||getClass()!=o.getClass()) return false;
52     Patient<T> patient=(Patient<T>)o;
53     return Objects.equals(id,patient.id);
54 }
55 @Override
56 public int hashCode(){
57     return Objects.hash(id);
58 }
59 public static void main(String[] args){
60     Scanner sc=new Scanner(System.in);
61     //(c) Store patient objects using an ArrayList
62     List<Patient<String>> patientList=new ArrayList<>();
63     //(d) Remove duplicate Patient IDs using HashSet
64     Set<String> uniquePatientIds=new HashSet<>();
65     //(e) Maintain a Mapping that maps Patient IDs with diagnosis de
66     Map<String,String> patientDiagnosisMap=new HashMap<>();
67     //(a) Read patient ID, name,age,gender,department,blood group,
68     if(sc.hasNextInt()){
69         int n=sc.nextInt();
70         sc.nextLine(); //consume newline
71         for(int i=0;i<n;i++){
72             String id=sc.nextLine().trim();
```

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JAVA PROGRAMMING

Vidhushini T S 1923110619C

```
71     for(int i=0;i<n;i++){
72         String id=sc.nextLine().trim();
73         String name=sc.nextLine();
74         int age=Integer.parseInt(sc.nextLine().trim());
75         String gender=sc.nextLine();
76         String department=sc.nextLine();
77         String bloodGroup=sc.nextLine();
78         String contactNumber=sc.nextLine();
79         String diagnosis=sc.nextLine();
80         //Deduplication check using HashSet(id)
81         if(!uniquePatientIds.contains(id)){
82             uniquePatientIds.add(id);
83             Patient<String> p=new Patient<>(id, name, age, ge
84             patientList.add(p);
85             patientDiagnosisMap.put(id,diagnosis);
86         }
87     }
88     //(f) Display all patient records using Iterator
89     System.out.println("== ALL UNIQUE PATIENT RECORDS ==");
90     Iterator<Patient<String>> iterator=patientList.iterator();
91     while(iterator.hasNext()){
92         System.out.println(iterator.next());
93     }
94     //(h) Produce department wise patient reports
95     System.out.println("\n== DEPARTMENT WISE PATIENT REPORT==");
96     Map<String,List<Patient<String>>> deptReport=new HashMap<>();
97     for(Patient<String> p:patientList){
98         deptReport.computeIfAbsent(p.getDepartment(),k->new Arrayl
99     }
100     for(Map.Entry<String,List<Patient<String>>> entry:deptReport.e
101         System.out.println("Department:"+entry.getKey());
102         for(Patient<String>p:entry.getValue()){
103             System.out.println(">">p.getName()+"(ID:">p.getId()+"
104         }
105     }
106     //(h) discussion ad collection interfaces (printed in output)
107     System.out.println("\n==ADVANTAGES OF COLLECTION INTERFACES==
108
```

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